RESEARCH PROPOSAL

Simulated Healthcare Claims Analytics:

Exploring Cost, Utilization, and Quality Metrics

October 2025

---

1. Abstract

This proposal outlines a simulated healthcare claims analytics project designed to examine cost drivers, utilization patterns, and quality indicators across a synthetic patient population. Using Python-based data simulation and machine learning, the project aims to surface actionable insights into healthcare spending efficiency, high-cost claimant identification, and outcome quality. The findings will demonstrate how AI and advanced analytics can be applied to address real-world challenges in healthcare operations, payer strategy, and policy design.

2. Background & Motivation

Healthcare expenditure in the United States exceeds $4 trillion annually, yet outcomes remain inconsistent across populations and providers. Claims data—encompassing medical, pharmacy, and behavioral health—represents one of the richest sources of information available to payers, employers, and policymakers. Yet this data is often underutilized due to its complexity, volume, and sensitivity.

Advances in machine learning and AI now enable analysts to move beyond descriptive statistics toward predictive and prescriptive insights. This project was motivated by a desire to explore how these techniques can be applied to healthcare claims to: (1) identify cost inefficiencies, (2) flag high-utilization individuals for care management, and (3) assess quality-of-care indicators at the cohort level.

Because real claims data carries significant privacy constraints, this project constructs a realistic synthetic dataset that mirrors the statistical properties of actual claims, enabling rigorous analysis without HIPAA concerns.

3. Research Objectives

- Simulate a realistic healthcare claims dataset including member demographics, diagnoses, procedures, and costs.
- Conduct descriptive analysis of cost and utilization metrics segmented by age, condition, and plan type.
- Develop predictive models (regression and classification) to forecast high-cost claimants.
- Evaluate quality metrics including preventive care adherence, readmission rates, and ER utilization.
- Visualize findings in dashboards and generate structured reports to communicate insights to stakeholders.

4. Methodology

4.1 Data Simulation

A synthetic dataset of 10,000 members will be generated using Python's NumPy and Faker libraries. Each record will include: member ID, age, gender, chronic condition flags (diabetes, hypertension, COPD, etc.), plan type, claims history (inpatient, outpatient, ER, pharmacy), total paid amounts, and quality flags.

4.2 Cost Analysis

Per-member-per-month (PMPM) cost will be calculated and segmented by plan type, condition burden, and age cohort. Pareto analysis will identify the top 10% of claimants driving ~70% of costs. Cost trend analysis will compare year-over-year changes in spending categories.

4.3 Utilization Analysis

Utilization metrics will include: ER visits per 1,000 members, inpatient admissions per 1,000, average length of stay, and pharmacy fill rates. Cohort comparisons will highlight where utilization is misaligned with clinical need.

4.4 Quality Metrics

Quality indicators will be modeled after HEDIS measures, including diabetic HbA1c monitoring rates, mammography screening rates, and 30-day readmission rates. A composite quality score will be calculated per cohort.

4.5 AI & Predictive Modeling

Scikit-learn will be used to build: (1) a gradient boosting regression model to predict next-year total cost, and (2) a binary classifier to identify high-cost claimants. SHAP values will be computed to explain model predictions and support clinical interpretability.

5. Expected Deliverables

Deliverable	Description
Research Proposal	This document outlining objectives, methodology, and timeline
Python Codebase	Data simulation, EDA, modeling, and visualization scripts
Final Analytical Report	Full written analysis with charts, findings, and recommendations
Presentation Deck	Executive-facing PPTX summarizing key insights and models

6. Project Timeline

Phase	Timeline	Key Activities
Phase 1	Weeks 1–2	Data simulation & EDA
Phase 2	Weeks 3–4	Cost & utilization analysis
Phase 3	Weeks 5–6	Quality metrics & model building
Phase 4	Week 7	Visualization & reporting
Phase 5	Week 8	Final review & presentation

7. Significance & Impact

This project bridges the gap between academic data science and real-world healthcare analytics. The skills developed—synthetic data generation, cost modeling, machine learning, and stakeholder communication—are directly transferable to roles in health insurance, health tech, consulting, and clinical operations.

By working through the full analytics lifecycle on healthcare claims, this project demonstrates the potential of AI-assisted tools to reduce costs, improve care coordination, and drive quality improvement at scale.

8. References

- Centers for Medicare & Medicaid Services (CMS). (2024). National Health Expenditure Data.
- NCQA. (2024). HEDIS Measures and Technical Resources.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.
- Lundberg, S. & Lee, S. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.